

Analysis of the TOP Framework UX Survey Results

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1 Introduction

This script processes and analyses data collected from an online survey evaluating user experience (UX) with the TOP Framework [1,2]. The User Experience Questionnaire (UEQ, version 11) [3] was used to assess the user experience. This includes steps for data preprocessing, calculation of summary statistics for UX-related scales, and generation of tables and visualizations to facilitate interpretation of the survey findings. The script is intended for research and reporting purposes, using survey data provided in CSV format.

The analysis supports understanding of user perceptions and experiences with the TOP Framework, providing insights for further development and improvement.

Analysis were performed with R version 4.3.3 (2024-02-29) [4].

2 Results

Table 1: Summary statistics of the TOP Framework UX survey results. Data collected from an online survey evaluating user experience with the TOP Framework.

Characteristic	N = 11	95% CI
scales		
attractiveness		1.07, 1.93
Mean (SD)	1.50 (0.64)	
Min - Max	0.50 - 2.50	
Median (Q1, Q3)	1.33 (1.17, 2.00)	
perspicuity		0.71, 1.70
Mean (SD)	1.20 (0.73)	
Min - Max	-0.25 - 2.25	
Median (Q1, Q3)	1.25 (0.75, 1.75)	
efficiency		0.93, 1.94
Mean (SD)	1.43 (0.75)	
Min - Max	0.25 - 2.25	
Median (Q1, Q3)	1.50 (0.75, 2.25)	
dependability		0.99, 2.03
Mean (SD)	1.51 (0.78)	
Min - Max	0.33 - 2.50	
Median (Q1, Q3)	1.25 (0.75, 2.25)	

Abbreviation: CI = Confidence Interval

Characteristic	N = 11	95% CI
stimulation		1.20, 2.21
Mean (SD)	1.70 (0.76)	
Min - Max	0.00 - 2.50	
Median (Q1, Q3)	2.00 (1.50, 2.25)	
novelty		0.29, 1.89
Mean (SD)	1.09 (1.20)	
Min - Max	-0.75 - 2.75	
Median (Q1, Q3)	1.50 (-0.25, 2.00)	
grouped scales		
attractiveness quality		1.07, 1.93
Mean (SD)	1.50 (0.64)	
Min - Max	0.50 - 2.50	
Median (Q1, Q3)	1.33 (1.17, 2.00)	
pragmatic quality		1.00, 1.77
Mean (SD)	1.38 (0.57)	
Min - Max	0.28 - 2.17	
Median (Q1, Q3)	1.42 (1.08, 1.92)	
hedonic quality		0.76, 2.03
Mean (SD)	1.40 (0.94)	
Min - Max	-0.38 - 2.63	
Median (Q1, Q3)	1.75 (0.63, 2.00)	

Abbreviation: CI = Confidence Interval

Table 2: t-test results for the TOP Framework UX survey scales.

name	n	estimate	alternative	conf.low	conf.high	p	p.signif
attractiveness	11	1.500	two.sided	1.072	1.928	0.004	**
dependability	11	1.508	two.sided	0.985	2.030	0.013	*
efficiency	11	1.432	two.sided	0.927	1.936	0.019	*
novelty	11	1.091	two.sided	0.288	1.894	0.438	ns
perspicuity	11	1.205	two.sided	0.713	1.696	0.097	ns

name	n	estimate	alternative	conf.low	conf.high	p	p.signif
stimulation	11	1.705	two.sided	1.196	2.213	0.003	**

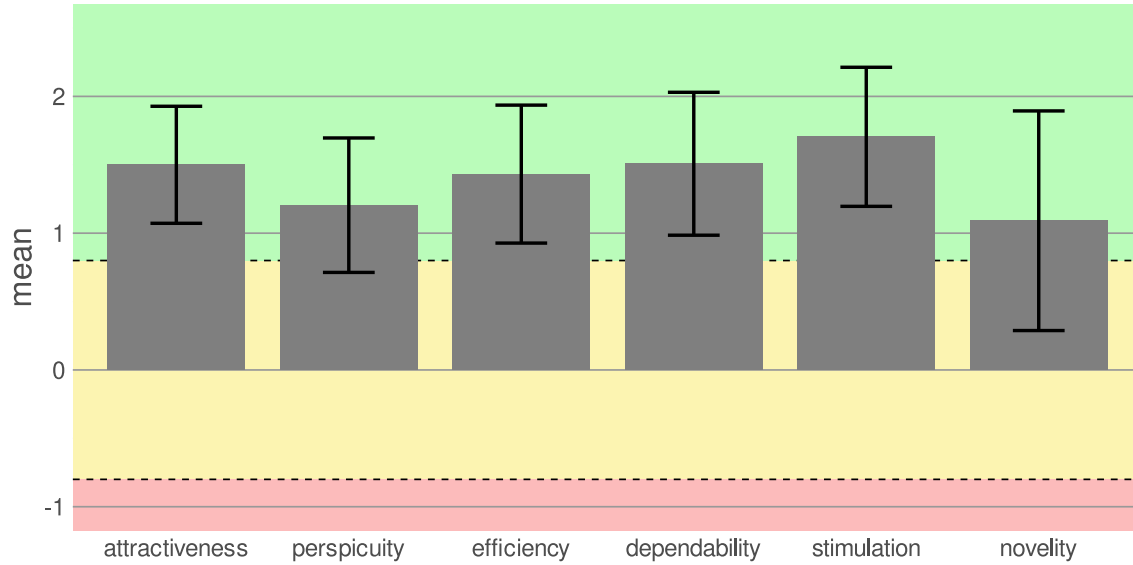


Figure 1: Mean values of the UX scales with confidence intervals. The scales are colour-coded based on their performance: red for poor, yellow for acceptable, and green for good.

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